Muhan Li

778-680-8519 | muhanli.work@gmail.com | linkedin.com/in/muhan-li | github.com/muhan-alan-li

EDUCATION

University of British Columbia Sep 2020 - May 2026
BASc. Computer Engineering – GPA: 3.8 *Vancouver, BC*

EXPERIENCE

Software Engineer Intern Sep 2025 - Apr 2026
Apera AI *Vancouver, BC*

- Shipped self-serve asset deployment with live status tracking (REST API, React/TypeScript), replacing a manual support-ticket workflow and cutting deployment time from ~12hrs to ~5min
- Extended Python/Redis DAG job scheduler with containerized execution and job caching, isolating deployment jobs and preventing redundant runs
- Added support for a new robot class, modeling joint limits and bin-picking behavior, enabling sales teams to validate product fit for prospective customer through simulation of production environment
- Implemented cross-user content sharing with recursive object copying and DynamoDB transactional writes, preserving nested object relationships and preventing dangling-reference failures
- Built continuous collision detection using spatial hashing in Three.js, maintaining 60+ FPS through memoization

Software Developer Co-op May 2025 - Aug 2025
Natural Resources Canada *Vancouver, BC*

- Ported the ANSE simulation engine from Windows-only .NET to a cross-platform C# runtime, replacing incompatible dependencies and refactoring platform-specific code to support Linux and macOS
- Shipped a C# transformation library that normalized multiple inconsistent Excel schemas into ANSE's engine-specific text format, centralizing format-specific conversion logic
- Automated ANSE build and deployment with GitHub CI/CD and containerized environments with Docker

Software Developer Co-op Jan 2023 - Aug 2023
UBC Cloud Innovation Center *Vancouver, BC*

- Delivered two full-stack AWS applications end-to-end for internal and external clients, translating requirements into serverless architectures, implementation plans, and deployed software
- Built a video-processing pipeline with Kinesis Video Streams, Rekognition, and Lambda for facial anonymization
- Shipped an anonymous drug-checking platform with React and AWS CDK, using Cognito, DynamoDB, API Gateway, and JWT-based authorization for anonymous and admin workflows

PROJECTS

Dune Imperium | *React, Typescript, Fastify, WebSockets* Sep 2026 - Present

- Built a 3–4 player multiplayer board game with server-driven state synchronization across connected clients.
- Developed a server-authoritative rules engine to validate actions and manage game progression and resources

Anteater | *RAG, WebLLM* Jul 2026 - Aug 2026

- Built an offline-first RAG pipeline for Canadian student loan information, crawling and cleaning government content, generating overlapping sentence-aligned chunks, and producing a static vector index
- Configured resource-aware model deployment with WebGPU checks, ensuring safe fallback without webLLM

SKILLS

Languages: TypeScript, Python, C++, C#, JavaScript, HTML/CSS, SQL
Frameworks: React, Node.js, Nest.js, GraphQL, RESTAPIs, .NET, Flask, Three.js, NoSQL, Sass
Tools: Agile, Claude, Git, Docker, AWS, Jira, Ubuntu, Terraform, Cloudwatch